

Chapter 14.7

Pytorch Transfer Learning Script for CNN Models

தமிழில்

Less than 30 Mins

AI COACH JOHN

CEO & FOUNDER

Agenda at a glance

1. Pytorch Script
2. **Pytorch?** *Misconceptions Vs Reality*
3. **Top 4 Reasons** on why do people use PyTorch a lot?
4. Brief History about Pytorch
5. Comparison based on 7 Aspects between **Pytorch** Vs **Tensorflow/Keras**



The Misconception

“We move to PyTorch because most **production-ready transfer learning libraries** are in PyTorch”



The Reality

Both TensorFlow and PyTorch **fully support** transfer learning and production.
You do **NOT need to switch to PyTorch** to use **pretrained models**.

Then why do people use PyTorch a lot?



Top 4 Reasons on why do people use PyTorch a lot?

1. Research & Experimentation

(Biggest Reason)

Examples:

- Vision Transformers
- CLIP
- SAM
- Diffusion models

👉 Why?

- Easier debugging
- More flexible
- Pythonic (feels natural)

⚙️ 2. Hugging Face ecosystem

👉 Major library:

- Hugging Face

👉 Most models:

- Native support in PyTorch first
- TensorFlow support comes later (sometimes)

🚀 3. Dynamic computation graph

👉 PyTorch:

- Builds graph on-the-fly

👉 TensorFlow (earlier versions):

- Static graph (harder to debug)

👉 This made PyTorch popular for:

- Research
- Custom architectures

🏭 4. Production? (Important clarification)

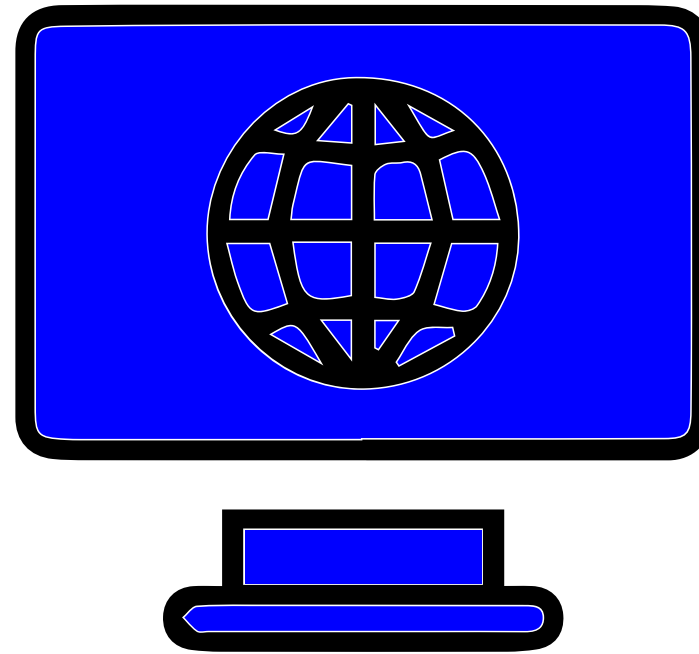
- TensorFlow → Strong production ecosystem (TensorFlow Serving, TFLite)
- PyTorch → Also production-ready (TorchServe, ONNX)

👉 Today:

Both are equally capable in production



Point 4



Brief History about Pytorch

Point 5

Comparison based on 7 Aspects between
Pytorch Vs Tensorflow/Keras

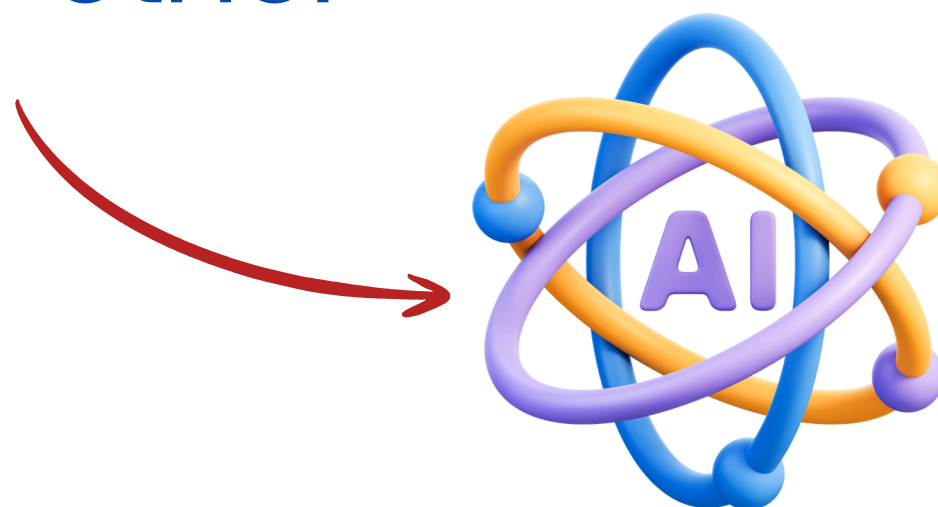
Comparison based on 7 Aspects between **Pytorch Vs Tensorflow/Keras**

No.	Aspect	Keras/TensorFlow	PyTorch
1	Learning curve	Easy ✅	Medium
2	Code style	High-level	Low-level
3	Control	Limited	Full control 🔥
4	Debugging	Harder	Easier
5	Research usage	Moderate	Very high
6	Interview preference	Medium	High 🔥
7	Flexibility	Medium	Very high

Update on
Pytorch MasterClass

Know one framework properly

To Convert to other
Frameworks



Overall Summary

1. Pytorch Script

2. **Pytorch?** *Misconceptions Vs Reality*

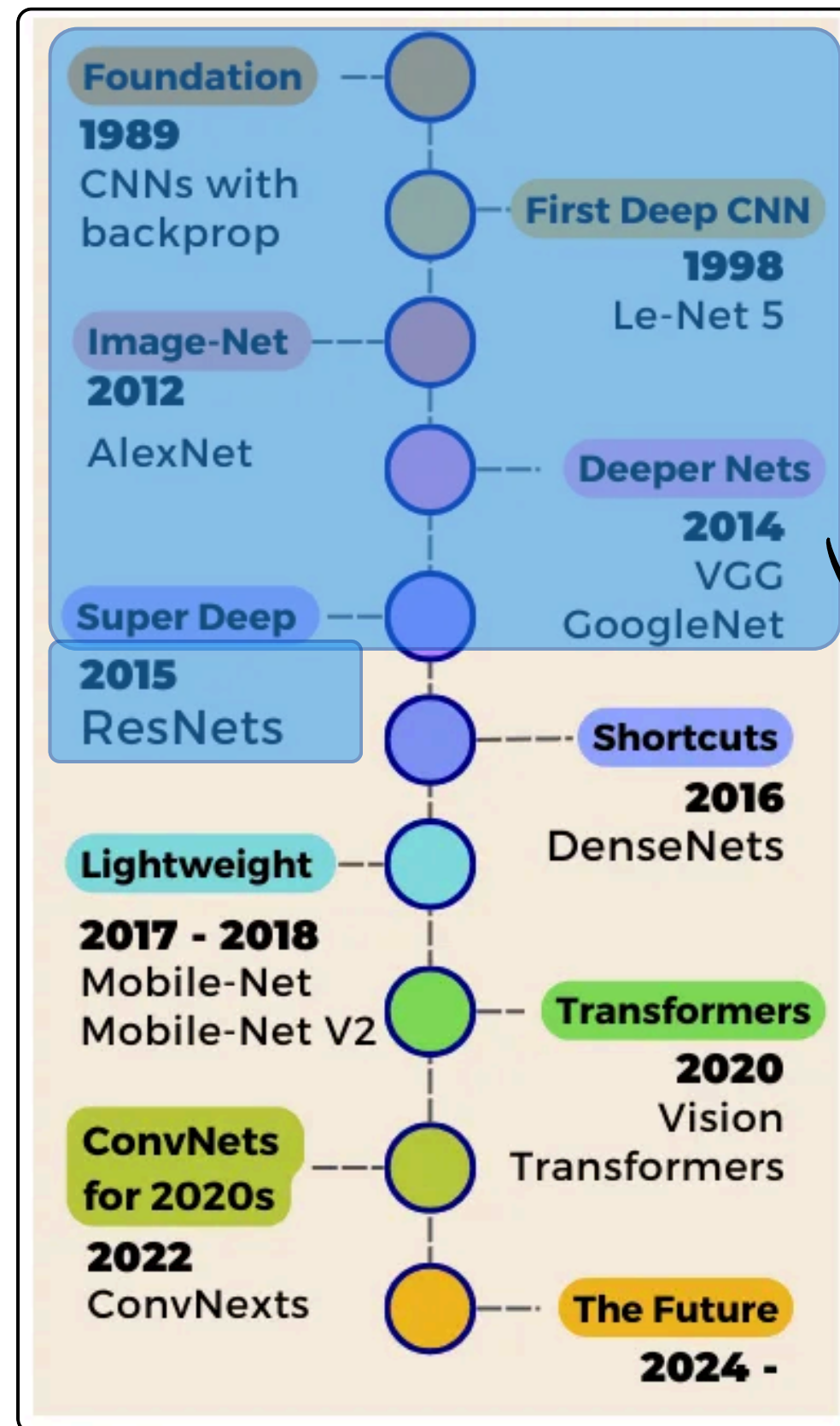
3. **Top 4 Reasons** on why do people use PyTorch a lot?

4. Brief History about Pytorch

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Where do we stand and the Plan?

Computer Vision Architectures: A Journey of Innovation



Era 1 → **CLASSIC CNN**
(1985 - 2015)

Foundations of Deep Learning

NEXT →

*We will now see
how GoogLeNet Model Works?*